

1. Python Program to Calculate the Area of a Triangle
2. Python Program to Make a Simple Calculator
3. Python Program to Convert Celsius To Fahrenheit
4. Python Program to Convert Kilometers to Miles
5. Calculate the speed of the car ,distance travelled and time taken is given.
6. Find nth power of a given number
7. Perform swapping of two numbers

A=3, B=4

After swapping A=4 and B=3

8. Write a program to display “Hello” if the number entered is divisible by 9 else print “Bye”
9. Write a program to calculate the electricity bill. Accept the number of units consumed by the user

Unit	Price
First 100 units	No charge
Next 100 units	Rs 5 per unit
After 200 units	Rs 10 per unit

For example, if the input unit is 350 then the total bill amount is Rs 2000

10. Write a program to check the last digit of a given number is divisible by 3
11. Write a program to accept the total mark (out of 500) of a student. Then calculate the percentage of marks. Calculate the grade based on the following criteria

Marks	Grade
≥ 80	A
≥ 70 and < 80	B
≥ 60 and < 70	C
≥ 50 and < 60	D
Below 50	Failed

12. Write a program to accept the cost price of a bike and display the road tax to be paid according to the following criteria

Price	Tax
>100000	15%
>50000 and <=100000	10%
<=50000	5%

13. Write a program to check whether given year is leap year or not
14. Write a program to accept a number from 1 to 7 and display the name of the day like 1 for Sunday 2 for Monday and so on.
15. Create a simple calculator. Perform the calculations by entering the operands and operator
16. Perform temperature conversion based on our requirement
Fahrenheit to Celsius: $C = (F - 32) \times (5/9)$
Celsius to Fahrenheit: $F = C \times (9/5) + 32$
17. Write a program to check whether the last digit of a given number is divisible by three or not
18. Write a program to find the lowest number out of two numbers.
19. Write a program to check whether a character is a vowel or not.
20. Accept the following from the user and calculate the percentage of attendance
Total number of working days
Total number of days absent
After calculating the percentage show that if the percentage is less than 75, then the student will not be able to attend the exam
21. A company decided to give bonuses to employees according to the following criteria:

Period of Service	Bonus
More than 10 years	10%
≥ 6 and ≤ 10	8%
Less than 6 years	5 %

Ask the user for their salary and years of service and print the net bonus amount
22. Accept three sides of a triangle and check whether it is an equilateral, isosceles, or scalene triangle
An equilateral triangle is a triangle in which all three sides are equal
A scalene triangle is a triangle that has three unequal sides
An isosceles triangle is a triangle that has any two sides equal
23. Accept the age, sex(M/F), number of days and display the wages accordingly

Age	Sex	Wage/Day
≥ 18 and < 30	M	700
	F	750
≥ 30 and ≤ 40	M	800
	F	850

If age does not fall in any range then display the following message: "Enter appropriate age"
24. Accept three different numbers then display the second largest number.

25. Accept the number of days from the user and calculate library charges

Till five days	Rs 2/day
Six to ten days	Rs 3/day
11 to 15 days	Rs 4/day
After 15 days	Rs 5/day

26. Python Program to Find the Largest Among Three Numbers

27. Check a given number is a perfect square.(16,25, 36... are perfect squares
 $4*4=16, 5*5=25, 6*6=36$)

28. Python Program to Solve Quadratic Equation.

LOOPS

29. Program to print all the numbers that are divisible by both 11 and 2 within range of 100-500

30. Create a program to print all the numbers in the range 10-50 excluding the multiples of 2 and 3

31. Check a given number is a perfect number or not(perfect number means the sum of factors equal to the same number $6=1+2+3$)

32. Python Program to Find Armstrong Number in an Interval

33. Python Program to Check Armstrong Number

34. Python Program to Find the Factorial of a Number

35. Python Program to Print the Fibonacci sequence less than a given number

36. Print first n terms of Fibonacci series

37. Python Program to Display the multiplication Table

38. Python Program to Check if a Number is Odd or Even

39. Python Program to Check if a Number is Positive, Negative or 0

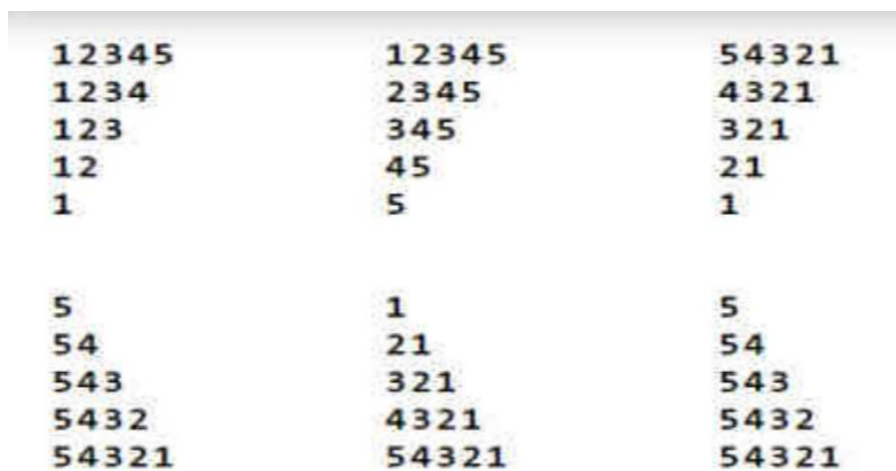
40. Python Program to Check Prime Number

41. Python Program to Print all Prime Numbers in an Interval

42. Python Program to Find the Sum of Natural Numbers

43. Find the sum of all the odd numbers below n

44. Find the sum of all even numbers and odd numbers separately up to a given number
45. Python Program to Find the Factors of a Number
46. Python Program to Find LCM
47. Convert a decimal number to binary
48. Convert a binary number to decimal
49. Print all the prime numbers within a range
50. Print all the perfect numbers within a range
51. Print all the Armstrong numbers within a range
52. Programs for printing pyramid patterns in Python
53. Program to print the diamond shape
54. Print an Inverted pyramid Star Pattern
55. Create following six patterns



56. Write a program to print the following pattern

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A
B C
D E F
G H I J

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FUNCTIONS

- 57. Python Program to Find Factorial of Number Using Recursion
- 58. Python Program to Display Fibonacci Sequence Using Recursion
- 59. Python Program to Find Sum of Natural Numbers Using Recursion
- 60. Python Program To Display Powers of 2 Using Anonymous Function
- 61. Python Program to Check Whether a String is Palindrome or Not

COLLECTIONS

- 62. Print the largest number in a list
- 63. Print smallest number in a list
- 64. Create a new list the does not contain the duplicate elements in a list
- 65. Create a program to split a list into two list. Here one list contains all the elements greater than 5 and other list contains all the elements less than 5
- 66. Create a program to enter two lists. Display True if the lists contains al least one common element.
- 67. Sort the elements in a list without using built in functions
- 68. Python program to interchange first and last elements in a list
- 69. Python program to swap two elements in a list
- 70. find length of list with of using len
- 71. Maximum of two numbers in Python without using max
- 72. Minimum of two numbers in Python without using min
- 73. Python program to check whether the string is Symmetrical or Palindrome
- 74. Reverse each words in a given String in Python
- 75. remove i'th character from string in Python
- 76. Find length of a string without using len
- 77. Python program to print even length words in a string
- 78. Check a given string is pangram or not(contains all the alphabets at least one time)
- 79. Check a given string is perfect pangram or not(contains all alphabets exactly one time)
- 80. Reverse of a number
- 81. Check two strings are anagram or not (eg: secure & rescue ,battle & tablet)

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82.      #Enter a string with more than  
        3 characters. Then find reverse
```

```

without changing first
#and last character
#eg:python--->pohtyn
83.      #* * * *
      # * * *
      #  * *
      #   *

```

84. Write a python function that takes two lists and returns True if they have at least one common number
85. Write a function to get a new string from a given string where “Is” has been added to the front.If the given string already begins with “Is” then return the string unchanged.
Eg: good morning
o/p : Is good morning
86. Print the largest word in a given string
87. Print largest words in a given string (needs to print if more than one word with same length)
88. Find the key in a dictionary that corresponding to largest number
Dic={'a':1,'b':4,'c':5}
M=max(dic,key=dic.get)
89. Check a given password is valid or not
A valid password must contain
 - more than 6 characters
 - at least one uppercase letter
 - at least one lowercase letter
 - at least one digit
 - at least one special character
86. Find the most frequent element in a given list
87. Check a given number is a happy number or unhappy number
88. Find the sum of $1 + \frac{2}{2!} + \frac{3}{3!} + \dots$ up to n terms
89. Check a given number is a perfect square
(Perfect squares examples 9 25 3*3=9 5*5=25)
90. Find the most repeating character in a string, also return the number of times that character repeated

91. Print first 10 prime numbers in reverse order
92. check the given number is Armstrong or not(it may contain any number of digits.)
93. Print the pattern
10
11 11
12 12 12
13 13 13 13
94. Print the key in a dictionary that corresponds to largest value

Create a class Shape. Inherit two classes circle and rectangle with the constructors. Find its area. Create the functions based on your requirements.

Break/continue

1. You need to input a string. Print all the characters in line by line until you get a vowel
2. Print the first vowel in a given string
3. Find

Check a number is Super Niven number

N=68040

68040 is divisible by 6,8,4,6+8,6+4,4+8,6+4+8

1.Print all the numbers within a range, these numbers must have at least one 2

Suppose range is 100 -200

output :102,112,120,121,122.....

2.Print all the numbers in a range,its digit sum exactly 3

3.Print all the numbers in a range, it contains atleast two 5s

